

Requirements Table — Community Lost & Found Matching Network

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall match newly logged 'Found Item' posts with active 'Lost Item' reports based on category, color tags, and location radius.	High	Pass: Match probability calculated and notification dispatched to potential owner. Fail: Incompatible item categories matched.	Core value proposition of the platform — automatically connecting finders and owners without manual searching.
FR-002	Functional	The system shall allow a Finder to post a Found Item report including at least one photo, a text description, category, and location.	High	Pass: Found Item appears in active listings with all four required fields populated. Fail: post is accepted with any required field missing.	Complete found-item data is a prerequisite for tag extraction (FR-004) and matching (FR-001).
FR-003	Functional	The system shall allow an Owner to submit a Lost Item report including a text description, category, last-seen location, and approximate date lost.	High	Pass: Lost Item report is stored, marked 'Active', and becomes eligible for matching. Fail: report is stored but not eligible for matching.	Lost reports are the other half of the matching pipeline; without them FR-001 has nothing to match against.
FR-004	Functional	The system shall automatically extract visual tags (e.g., color, object type) from uploaded photos and text tags from descriptions for both Found and Lost item posts.	Medium	Pass: at least one visual tag and one text tag are attached to a post within 5 seconds of submission. Fail: post has zero tags after submission completes.	Automated tagging feeds the matching algorithm (FR-001); without it, matching would require fully manual entry.
FR-005	Functional	The system shall require a claimant (Owner) to correctly answer the Finder's proof-of-ownership security question(s) before revealing the Finder's contact details.	High	Pass: contact details are shown only after a correct answer is submitted. Fail: contact details are visible without a correct answer.	Prevents fraudulent claims and protects finder privacy; directly implements the control required by NFR-001.
NFR-001	Nonfunctional	Claimants must answer private security questions set by the finder before item handover contact details are revealed.	High	Pass: benchmarking tests confirm target latency and security standards under simulated peak load.	Protects user privacy and prevents fraudulent handovers while keeping verification fast enough not to frustrate legitimate owners.
NFR-002	Nonfunctional	The system shall complete tag extraction and match calculation for a newly posted Found Item within 5 seconds under normal load (≤ 100 concurrent postings).	Medium	Pass: 95th percentile response time for tag-extraction + matching $\leq 5s$ in load testing. Fail: 95th percentile exceeds 5s.	Timely notifications increase the chance a Finder and Owner are reunited before the item is discarded or the post goes stale.

Actors: Finder, Owner, Community Admin

